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-- 📊 Revenue Leakage & Data Quality Analysis (Zomato Dataset)
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-- 🧹 1. DATA CLEANING
-- ==========================

-- Create database
CREATE DATABASE zomato;

-- Check total records
SELECT COUNT(*)
FROM zomato.zomato_dataset;

-- Preview dataset
SELECT *
FROM zomato.zomato_dataset
LIMIT 10;

-- Create cleaned table
CREATE TABLE zomato.clean_data AS
SELECT *
FROM zomato.zomato_dataset;

-- Convert invalid ratings ('NEW', '-') to NULL
UPDATE zomato.clean_data
SET Delivery_Rating = NULL
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WHERE Delivery_Rating IN ('NEW', '-');

-- Convert 0 ratings to NULL (invalid values)
SET SQL_SAFE_UPDATES = 0;

UPDATE zomato.clean_data
SET Dining_Rating = NULL
WHERE Dining_Rating = 0;

UPDATE zomato.clean_data
SET Delivery_Rating = NULL
WHERE Delivery_Rating = 0;

-- Validate cleaning
SELECT
  COUNT(*) AS total_rows,
  COUNT(Dining_Rating) AS valid_dining_ratings,
  COUNT(Delivery_Rating) AS valid_delivery_ratings
FROM zomato.clean_data;

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-- 📊 2. REVENUE ANALYSIS
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-- Revenue by city (identify top markets)
SELECT
  City,
  SUM(Prices) AS total_revenue
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FROM zomato.clean_data
GROUP BY City
ORDER BY total_revenue DESC;

-- Top 3 revenue-generating cities
SELECT
  City,
  SUM(Prices) AS total_revenue
FROM zomato.clean_data
GROUP BY City
ORDER BY total_revenue DESC
LIMIT 3;

-- Analyze revenue drivers (volume vs price)
SELECT
  City,
  COUNT(*) AS total_orders,
  AVG(Prices) AS avg_price,
  SUM(Prices) AS total_revenue
FROM zomato.clean_data
GROUP BY City
ORDER BY total_revenue DESC
LIMIT 8;

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-- 💸 3. REVENUE LEAKAGE ANALYSIS
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-- Estimate discount impact (~10% assumption)
SELECT
  City,
  SUM(Prices) AS total_revenue,
  SUM(Prices * 0.1) AS estimated_discount_loss
FROM zomato.clean_data
GROUP BY City
ORDER BY estimated_discount_loss DESC;

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-- 🔍 4. DATA QUALITY CHECKS
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-- Fix column name for consistency
ALTER TABLE zomato.clean_data
CHANGE `Dining Votes` Dining_Votes INT;

-- Detect suspicious ratings (high rating, low votes)
SELECT
  Place_Name,
  Dining_Rating,
  Dining_Votes
FROM zomato.clean_data
WHERE Dining_Rating >= 4.5
AND Dining_Votes < 50;

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-- 🧠 5. ADVANCED ANALYSIS
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-- Top 3 restaurants by revenue per city (window function)
SELECT *
FROM (
   SELECT
      City,
      Place_Name,
      SUM(Prices) AS revenue,
      RANK() OVER (PARTITION BY City ORDER BY SUM(Prices) DESC) AS rnk
   FROM zomato.clean_data
   GROUP BY City, Place_Name
) t
WHERE rnk <= 3;
```